

Counting Fruits with State-of-the-Art Multi-View-Stereo Approaches

Bachelor Thesis in Computer Science

presented by

Alexander Link

born on 29.08.2004 in Bamberg

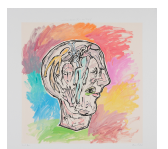
submitted to

Department of Computer Science
Professorship for Cognitive Computer Vision
Friedrich-Alexander-Universität Erlangen-Nürnberg

Academic Supervision: Prof. Dr. Bernhard Egger

Begin date: 01.05.2026

End date: 01.10.2026



Abstract

Automated fruit counting from RGB images supports yield estimation in precision agriculture. Existing approaches such as FruitneRF achieve this through scene-specific training of a neural radiance field, which requires time-consuming optimization per scene. In this work, we investigate whether the feed-forward multi-view stereo model VGGT can replace this training by predicting a 3D reconstruction in a single forward pass, thereby significantly reducing runtime. To this end, we assign 2D instance masks to the views based on the 3D representation predicted by VGGT across all views and count the fruits using clustering with preceded partial instance association.

Evaluation,
Ergebnisse

Contents

1	Introduction	1
2	Related Work	3
2.1	Fruit Counting and Yield Estimation	3
2.2	3D-Reconstruction	4
2.3	Preparation of the Bibliography	4
2.4	Citing a Paper	4
3	Method	7
3.1	Tense	7
3.2	I, We, Passive voice	7
3.3	Equations	7
3.4	Figures	7
3.5	Tables	8
4	Implementation	9
4.1	Listings	9
5	Results	11
6	Conclusions	13
6.1	Tips	13

CONTENTS

Chapter 1

Introduction

The automatic counting of fruit from images is a key task in precision agriculture, where accurate yield estimates support harvest planning and resource allocation. Reconstructing a fruit tree scene in 3D from a set of RGB images and counting the fruit it contains is appealing because it requires only inexpensive and widely available camera hardware, rather than specialized depth sensors.

citation

Recent Work such as FruitNeRF [5] show that this is possible by training a Neural Radiance Field [6] per scene and counting the fruits in the resulting volumetric representation. This approach is effective but requires a time-consuming scene-specific optimization. In this work, we instead investigate whether a feed-forward multi-view stereo model, VGGT [8], can replace scene-by-scene training. Instead of optimizing a representation of each scene, VGGT predicts a 3D reconstruction in a single forward pass. This promises a significant reduction in runtime, but raises the central question of whether the resulting geometry is sufficiently accurate for reliable counting.

Counting fruits in this scenario is challenging for several reasons: Fruits are often obscured by foliage and each other, the appearance of a scene varies with changing lighting, and the geometric complexity of a tree canopy makes a consistent reconstruction difficult. To reduce occlusion and the risk of missing apples, a scene should ideally be captured from all sides, but the number of views is limited by hardware constraints such as available memory. The cameras are therefore sparsely distributed across a full 360-degree range, and widely spaced views often capture completely different regions of the tree. This makes it difficult to establish correspondences between views, so that individual seen only by a few cameras are at risk of being fragmented in the reconstruction.

This results in two types of opposing types of errors. Individual apples observed by only a few cameras yield unstable correspondences and are easily split into multiple detections, while dense clusters of apples seen by many cameras may merge into a single detection. A single set of static parameters can therefore rarely resolve both cases optimally at the same time.

Our approach maps semantic information in the form of 2D instance masks, obtained using SAM3 [1], across views via the 3D representation predicted by VGGT. The fruits are then counted by combining a clustering step with a preceding partial instance association that supports the clustering.

We evaluate the approach in different ways on multiple datasets

datasets,
performance

Our key findings suggest that the scene complexity is the decisive factor for counting accuracy, since the quality of the VGGT reconstruction varies significantly depending on lighting, depth complexity, and whether the input is synthetic or real.

key findings

Chapter 2

Related Work

2.1 Fruit Counting and Yield Estimation

Approaches to fruit counting can be distinguished based on whether they operate in the image plane or in a tree-dimensional representation of the scene. Traditional image-based pipelines are divided into two subtasks: the detection of fruits in the image and their counting across multiple views [haeni2020]. While detection used to rely on handcrafted color and shape features, it is now dominated by deep neural networks [miranda2023]. For example, YOLO-based detectors achieve high accuracies on single images [ballenaruiz2025, rameshchandra2025]. However, such methods count fruits on a per-frame basis and lack consistent instance tracking across multiple views, meaning that the same fruit can be counted multiple times. It is precisely this lack of cross-view tracking that is the problem addressed by the 3D-based approach pursued in this work.

This double counting is the central challenge in counting across image sequences. Liu et al. [liu2018] introduced the first complete pipeline that tracks fruits across consecutive images and uses Structure-from-Motion (SfM) to correct double counting. In a subsequent paper, Liu et al. [liu2019] combined detection with a semantic SfM reconstruction and assigned detections to one another using the Hungarian algorithm, using only a single camera. The association using the Hungarian algorithm is also adopted in the present work; however, there it operates on densely sampled video sequences with short frame intervals, whereas the scenario considered here is based on widely spaced 360-degree views with weak correspondences between the views.

To structurally avoid double counting, later studies shifted the counting entirely into three-dimensional space. Häni et al. [haeni2020] reconstructed rows of trees from both sides and merged the counts by combining the 3D models. Gené-Molá et al. [genemola2020] combined Mask R-CNN instance segmentation with an SfM point cloud, projecting the 2D masks into 3D space, and filtered out false detections using a support vector machine. This approach is the closest in structure to the present work, as it maps 3D instance masks to views via a 3D representation. However, it relies on an SfM reconstruction, which takes several hundred minutes per scene to complete and is therefore not suitable for a fast run, precisely the limitation that this work seeks to overcome by using a feed-forward model. FruitNeRF [5] fits into this trend as the approach that fully integrates counting in to a neural scene representation, thereby overcoming the previous separation of reconstruction and segmentation. However, since the NeRF representation used must be optimized per scene, the computational cost remains

high. This limitation forms the immediate starting point of this work. More recent works address this by replacing per-scene optimization with faster representations: FruitGaussian [hong2025] and CountingFruit [li2025] use 3D Gaussian splatting to significantly speed up reconstruction, but they remain within the framework of scene-specific optimization.

2.2 3D-Reconstruction

2.3 Preparation of the Bibliography

To add references, first add the corresponding BibTeX entries in the file `bibliography.bib`.

Journal papers use the `@article` entry, which needs at least:

- **title:** Title of the paper
- **author:** Separate authors with "and" and use comma to separate last from first names
- **journal:** Name of the journal.
- **volume:** Volume number. Each year a new volume is created.
- **number:** Issue number within a year. The journal might have multiple issues in a year.
- **pages:** Page range.
- **year:** Year of appearance.

Conference papers use the `@inproceedings` entry, which needs at least:

- **title:** Title of the paper.
- **author:** Separate authors with "and" and use comma to separate last from first names
- **booktitle:** Name of the conference.
- **pages:** Page range.
- **year:** Year of appearance.

2.4 Citing a Paper

To cite a paper in the LaTeX document, use the `cite` command. You can use indirect quotations at the end of a sentence [10].

Or you directly reference an author, such as: Weyrich et al. [10] described a method. If a paper has more than two authors, it is common to use “et al.”, which means “and colleagues”.

There are also several commands that can help when citing a paper, e.g., when you want to cite in a given paper in a text: Weyrich et al. [10].

`cite` Cite only square brackets: [10]

`cite` Cite multiple sources: [3, 7]

`textcite` Cite textual: Egger et al. [2]

`textcite` Cite multiple sources: Weyrich et al. [10] and Harrer et al. [4]

`citeyear` Cite the year: 2023.

Chapter 3

Method

3.1 Tense

Mind your use of present and past tense. Reports are typically written in present tense, except for the related work and conclusion section, which are usually in past tense. If you describe what past authors did, use past tense. If you describe what an algorithm does, which still works the same way today, use present tense.

3.2 I, We, Passive voice

It is very uncommon to write “I”. Instead, either avoid using personal pronouns by switching to a passive voice, or use “we” instead. When reading the word “we”, you can think of this as “the reader and me”.

3.3 Equations

For equations, the `align` command is recommended, since it allows you to align equations at a specific location, such as at the `=` sign.

$$y = f(x) \tag{3.1}$$

$$\boldsymbol{f} = \boldsymbol{m} \cdot \boldsymbol{a}^\top \tag{3.2}$$

Equations are referenced using `eqref`, for example Equation (3.1) and Equation (3.2). See `macros.tex` for a range of convenient macros, e.g., vector $\boldsymbol{x}$ and matrix $\boldsymbol{M}$ symbols

3.4 Figures

Figure placement is best done with the default `figure` environment. To reference a figure use `figref`, for example Figure 3.1. If possible, always create figures as vector graphics. For example an overview / pipeline graphic of your method should always be a vector graphic. Vector graphics allow zooming into figures without a loss in quality of the figure. Vector graphics can be exported in PDF format from common editing programs, which allows including them in Latex with `includegraphics`. Raster

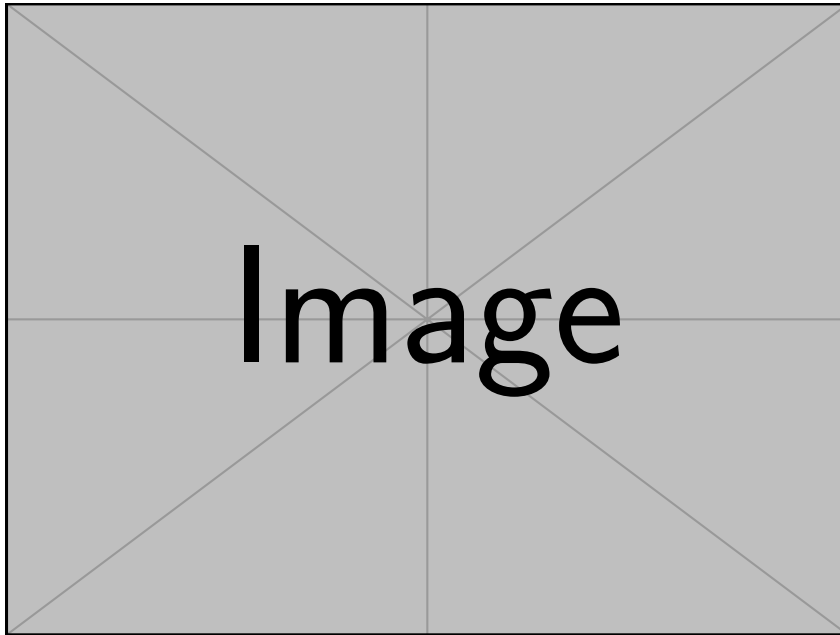


Figure 3.1: Caption of the figure. Always cite the source of the image in the caption, unless you created the image yourself.

	Metric 1↑	Metric 2↑	Metric 3↓	Metric 4↓
Method 1	23.1388	0.8611	0.2479	0.1846
Method 2	28.3813	0.9353	0.1831	0.1047
Ours	26.2468	0.9257	0.1509	0.0915

Table 3.1: Caption of the Table

images should only be used in your thesis when an image is shown, that is naturally rendered at a given resolution, e.g., a result image of a rendering algorithm.

3.5 Tables

Table placement is best done with the default `table` environment. To reference a table use `tblref`, for example Table 3.1.

Chapter 4

Implementation

While the previous section aims to describe your method in a platform independent way, this section presents all the details to reproduce your work.

4.1 Listings

Not all theses require the inclusion of source code. If it is necessary, Code placement is best done with the `lstlisting` environment.

```
1 # Comment
2 def function(parameter):
3     print("This is an example Python listing")
4     return True
```

Listing 4.1: Caption of the Python Listing

```
1 // Comment
2 bool function(int parameter){
3     std::cout << "This is an example C++ listing" << std::endl;
4     return true;
5 }
```

Listing 4.2: Caption of the C++ Listing

To reference a listing use `lstref`, for example Listing 4.1.

Chapter 5

Results

Describe your qualitative and quantitative evaluations and present your results. Discuss the limitations of your work.

Chapter 6

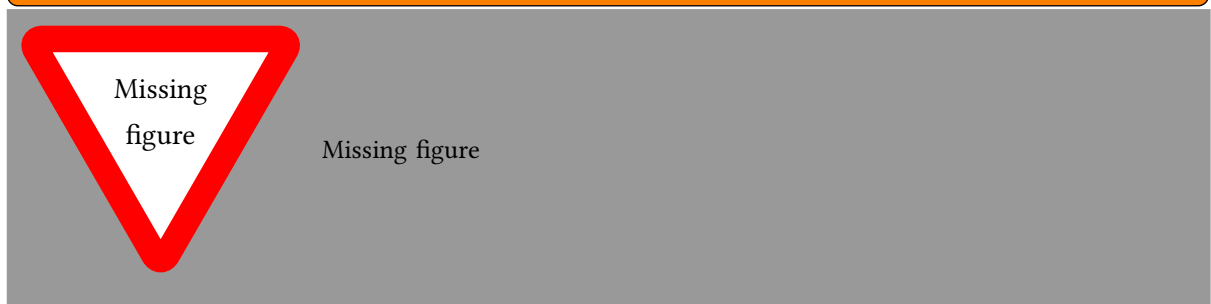
Conclusions

Give a brief summary of your work and your contributions. Afterwards, briefly outline potential opportunities for future work.

6.1 Tips

As simple package to keep track of unfinished areas of the thesis is the `todonotes` package.

Inline TODO



This is
a simple
TODO
note

Create a list of remaining TODOs with `listoftodos`. Don't forget to remove the `listoftodos` before submitting your thesis.

Todo list

Evaluation, Ergebnisse	iii
citation	1
datasets, performance	1
key findings	1
This is a simple TODO note	13
Inline TODO	13
Figure: Missing figure	13

List of Figures

3.1 Caption of the Figure as listed in listoffigures 8

LIST OF FIGURES

List of Tables

3.1 Caption of the Table as listed in listoftables 8

LIST OF TABLES

List of Listings

4.1	Caption of the Python Listing	9
4.2	Caption of the C++ Listing	9

LIST OF LISTINGS

Bibliography

- [1] Nicolas Carion et al. *SAM 3: Segment Anything with Concepts*. 2026. arXiv: 2511.16719 [cs.CV]. URL: <https://arxiv.org/abs/2511.16719>.
- [2] Bernhard Egger, William AP Smith, Ayush Tewari, Stefanie Wuhler, Michael Zollhoefer, Thabo Beeler, Florian Bernard, Timo Bolkart, Adam Kortylewski, Sami Romdhani, et al. “3d morphable face models—past, present, and future”. In: *ACM Transactions on Graphics (ToG)* 39.5 (2020), pp. 1–38.
- [3] Tobias Günther, Kai Rohmer, and Thorsten Grosch. “GPU-accelerated Interactive Material Aging”. In: *Vision, Modeling and Visualization*. Ed. by Michael Goesele, Thorsten Grosch, Holger Theisel, Klaus Toennies, and Bernhard Preim. The Eurographics Association, 2012, pp. 63–70. ISBN: 978-3-905673-95-1. DOI: 10.2312/PE/VMV/VMV12/063-070.
- [4] Mathias Harrer, Linus Franke, Laura Fink, Marc Stamminger, and Tim Weyrich. “Inovis: Instant Novel-View Synthesis”. In: *SIGGRAPH Asia Conference Papers*. Sydney, NSW, Australia: Association for Computing Machinery, Dec. 12–15, 2023. DOI: 10.1145/3610548.3618216. URL: <https://doi.org/10.1145/3610548.3618216>.
- [5] Lukas Meyer, Andreas Gilson, Ute Schmid, and Marc Stamminger. “FruitNeRF: A Unified Neural Radiance Field based Fruit Counting Framework”. In: *2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. 2024, pp. 1–8. DOI: 10.1109/IROS58592.2024.10802065.
- [6] Ben Mildenhall, Pratul P. Srinivasan, Matthew Tancik, Jonathan T. Barron, Ravi Ramamoorthi, and Ren Ng. “NeRF: representing scenes as neural radiance fields for view synthesis”. In: *Commun. ACM* 65.1 (Dec. 2021), pp. 99–106. ISSN: 0001-0782. DOI: 10.1145/3503250. URL: <https://doi.org/10.1145/3503250>.
- [7] Marc Stamminger and George Drettakis. “Perspective shadow maps”. In: *Proceedings of the 29th annual conference on Computer graphics and interactive techniques*. 2002, pp. 557–562.
- [8] Jianyuan Wang, Minghao Chen, Nikita Karaev, Andrea Vedaldi, Christian Rupprecht, and David Novotny. “VGGT: Visual Geometry Grounded Transformer”. In: *2025 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 2025, pp. 5294–5306. DOI: 10.1109/CVPR52734.2025.00499.
- [9] Maximilian Weiherer and Bernhard Egger. *From Zero to Hero: Convincing with Extremely Complicated Math*. 2023. arXiv: 2304.00399 [cs.CV]. URL: <https://arxiv.org/abs/2304.00399>.

BIBLIOGRAPHY

- [10] Tim Weyrich, Wojciech Matusik, Hanspeter Pfister, Bernd Bickel, Craig Donner, Chien Tu, Janet McAndless, Jinho Lee, Addy Ngan, Henrik Wann Jensen, et al. “Analysis of human faces using a measurement-based skin reflectance model”. In: *ACM Transactions on Graphics (ToG)* 25.3 (2006), pp. 1013–1024.

Declaration

I certify that I have written this thesis without outside help and without using sources other than those indicated, and that the thesis has not been submitted in the same or similar form to any other examination authority and has not been accepted by them as part of an examination. All statements that have been taken over verbatim or in spirit are marked as such.

During the preparation of this work, AI technologies were used to assist in the writing process. Specifically, <AI Tool 1 (Company, City, State, Country)> was used in order to check for grammar and style consistency, and <AI Tool 2 (Company, City, State, Country)> was used in order to assist with rephrasing and improving readability. After using these tools, the manuscript was carefully reviewed and the content was edited as needed. No tools or services were used for content generation. The content of this thesis was entirely created by myself, and I ensured that all material presented reflects my original work and research. I take full responsibility for the content of the thesis.

I agree that the thesis may be published and that it may be referred to in scientific publications.

The Friedrich-Alexander-Universität Erlangen-Nürnberg, represented by the Professorship for Cognitive Computer Vision, is granted a (non-exclusive) right to use this work and the programs created in connection with it.

Erlangen, 01.10.2026

(Alexander Link)